Assignment 4: ResNet with Spectrograms

Reference paper: Najeh, Lundberg, Kerrouche (2021). *Deep-Learning and Vibration-Based System for Wear Size Estimation of Railway Switches and Crossings.* Sensors, 21, 5217.

1. Background (read paper Sections 1–3 first)

Railway **Switches & Crossings (S&C)** wear out over time. Direct inspection (cameras, manual tools) is expensive. The paper proposes an **indirect** approach: install accelerometers inside the **point machine** and let a deep-learning model estimate the wear level from the **vibration signal** of a bogie passing over the S&C.

In this assignment you will work on the **middle section (S1S2)** of the test rig and classify the wear into four levels.

Target labels (4 classes): L0 (before grinding), L1 (1st grinding), L2 (2nd grinding), L3 (3rd grinding) (see Table 4 of the paper for the actual wear depths).

2. The data

- **Sampling rate:** 51.2 kHz (vibration), 1 Hz (speed).
- **Data:** Vibration converted to spectrograms

3. ResNet-18 on spectrogram images

Goal

Train a **ResNet-18** that classifies the wear level (L0/L1/L2/L3) from a **spectrogram images**.

Step 2: Build / load ResNet-18

Step 3: Train

- 60 % / 40 % split.
- ~20–30 epochs are usually enough when fine-tuning a pre-trained net.
- Track training and test accuracy per epoch.

Step 4: Evaluate

- Confusion matrix (4×4) compare with paper Fig. 14 (paper reports ≈ 94 % accuracy, better than LSTM ?).
- Plot a few spectrograms with their predicted vs true label.

What to hand in

- Notebook or script with spectrogram generation + ResNet fine-tuning.
- Final confusion matrix and overall accuracy.
- Short discussion (1 page): why does the image-based view help, compared with extracting hand-crafted features? (Hint: paper §4.4.)